Process & Decision Documentation

Project/Assignment Decisions:

For this assignment, I decided to structure the platformer game using a data driven approach with JSON and classes. Instead of hardcoding levels, I used a JSON format to define platforms, player start positions, and themes. This allowed me to separate game data from logic, making the code easier to organize and extend.

I also chose to break the program into multiple classes such as Platform, WorldLevel, and BlobPlayer, so each part of the game had a clear responsibility.

GenAI Documentation

Date Used: February 26th 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use:
- Help debug errors in HTML and JavaScript files
- Assist in structuring JSON level data correctly
- Improve code organization using classes
- Generate explanations for concepts like loading JSON and rendering levels
- Help format documentation such as the README and this process report

Summary of Interaction: I used GenAI in an interactive way to troubleshoot issues and improve my implementation. For example, I asked for help fixing errors caused by duplicate script tags and JSON formatting issues. I also requested suggestions for structuring levels, handling theme colors, and implementing features such as automatic text colour adjustment.

Human Decision Point: Although GenAI suggested solutions, I made the final decisions on:

- How to structure my files and classes
- What features to include in the game
- How levels should be designed and balanced
- Which parts of the generated code to keep or modify

Integrity & Verification Note: All AI generated suggestions were reviewed and tested manually. I ensured that

- The code runs correctly in my environment
- The logic matches course requirements
- The final implementation reflects my understanding

Scope of GenAI Use:

- Debugging assistance
- Code suggestions and explanations
- Documentation writing support

Limitations or Misfire: Some suggestions from GenAI did not directly match my project setup, such as incorrect assumptions about file structure or how JSON was being loaded. In some cases, I needed to adjust the provided code to fit my existing implementation. GenAI also cannot fully understand my project context without complete code, so I had to refine prompts and verify outputs.

Summary of Process (Human + Tool): The project was developed iteratively. I first implemented a basic platformer using p5.js, then refactored the code into classes. After that, I introduced JSON to define levels, which improved flexibility and scalability. When errors occurred, such as script loading issues or JSON formatting problems, I used GenAI to help identify possible causes and solutions. I then tested these solutions and integrated them into my code. The final result is a modular system where levels can be changed without modifying the core game logic.

Decision Points & Trade-offs: One major decision was whether to hardcode levels or use JSON. Using JSON required more setup but made the project more flexible. Another decision was organizing the code into classes. This increased complexity initially but made the code cleaner and easier to manage. I also chose to keep the game simple rather than adding advanced features, in order to focus on demonstrating the required concepts clearly.

Verification & Judgement: I tested the program by:

- Running multiple levels
- Checking player movement and collisions
- Ensuring JSON data loads correctly
- Verifying that themes and platforms display as expected

Limitations, Dead Ends, or Open Questions:

One limitation is that the game currently has basic mechanics and no advanced features such as enemies, animations, or scoring. I also encountered issues with loading JSON without a local server, which required additional setup.

Appendix

My prompt:

“Give me a new background with new x&y platforms n different blob colour or symbol for each level”

ChatGPT 5.2 - output #1:

“Here’s an updated version with **more levels**, **new backgrounds**, **different platform layouts (x, y, w, h)**, and **a distinct blob color + symbol per level** (I added `blobSymbol` and `blobStyle` so each level can visually differ).


```
{
  "schemaVersion": 1,
  "levels": [
    {
      "name": "Neon Alley",
      "gravity": 0.7,
      "jumpV": -11.2,
      "theme": {
        "bg": "#0B1020",
        "platform": "#2B3A67",
        "blob": "#00E5FF",
        "accent": "#FF2D95"
      },
      "blobStyle": "solid",
      "blobSymbol": "◆",
      "start": { "x": 70, "y": 240, "r": 24 },
      "platforms": [
        { "x": 0, "y": 336, "w": 640, "h": 36 },
```

```
    { "x": 90, "y": 270, "w": 120, "h": 12 },
    { "x": 250, "y": 220, "w": 90, "h": 12 },
    { "x": 380, "y": 170, "w": 140, "h": 12 },
    { "x": 560, "y": 250, "w": 70, "h": 12 }
  ]
},
{
  "name": "Candy Clouds",
  "gravity": 0.62,
  "jumpV": -10.6,
  "theme": {
    "bg": "#FFF1F7",
    "platform": "#FFC2D6",
    "blob": "#7C4DFF",
    "accent": "#00C853"
  },
  "blobStyle": "striped",
  "blobSymbol": "♥",
  "start": { "x": 60, "y": 210, "r": 26 },
  "platforms": [
    { "x": 0, "y": 334, "w": 640, "h": 36 },
    { "x": 70, "y": 260, "w": 110, "h": 12 },
    { "x": 220, "y": 200, "w": 120, "h": 12 },
    { "x": 390, "y": 240, "w": 80, "h": 12 },
    { "x": 500, "y": 170, "w": 120, "h": 12 },
    { "x": 590, "y": 260, "w": 50, "h": 12 }
```

```
  ]
},
{
  "name": "Forest Ruins",
  "gravity": 0.75,
  "jumpV": -12.0,
  "theme": {
    "bg": "#0F1F16",
    "platform": "#3D6B4F",
    "blob": "#FFD54F",
    "accent": "#A5D6A7"
  },
  "blobStyle": "glow",
  "blobSymbol": "☘",
  "start": { "x": 95, "y": 245, "r": 23 },
  "platforms": [
    { "x": 0, "y": 336, "w": 640, "h": 36 },
    { "x": 130, "y": 285, "w": 95, "h": 12 },
    { "x": 260, "y": 240, "w": 90, "h": 12 },
    { "x": 420, "y": 210, "w": 110, "h": 12 },
    { "x": 520, "y": 155, "w": 95, "h": 12 },
    { "x": 350, "y": 140, "w": 70, "h": 12 }
  ]
},
{
  "name": "Ice Circuit",
```

```
  "gravity": 0.58,
  "jumpV": -10.4,
  "theme": {
    "bg": "#E6F7FF",
    "platform": "#A7D8FF",
    "blob": "#1565C0",
    "accent": "#00B8D4"
  },
  "blobStyle": "outlined",
  "blobSymbol": "❄",
  "start": { "x": 80, "y": 230, "r": 25 },
  "platforms": [
    { "x": 0, "y": 334, "w": 640, "h": 36 },
    { "x": 120, "y": 270, "w": 80, "h": 12 },
    { "x": 240, "y": 230, "w": 85, "h": 12 },
    { "x": 360, "y": 255, "w": 90, "h": 12 },
    { "x": 470, "y": 210, "w": 110, "h": 12 },
    { "x": 560, "y": 170, "w": 70, "h": 12 }
  ]
},
{
  "name": "Volcano Vault",
  "gravity": 0.82,
  "jumpV": -12.6,
  "theme": {
    "bg": "#1B0B0B",
```

```
      "platform": "#6D2C2C",
      "blob": "#FF6D00",
      "accent": "#FFD600"
    },
    "blobStyle": "lava",
    "blobSymbol": "▲",
    "start": { "x": 65, "y": 245, "r": 22 },
    "platforms": [
      { "x": 0, "y": 336, "w": 640, "h": 36 },
      { "x": 80, "y": 290, "w": 100, "h": 12 },
      { "x": 210, "y": 235, "w": 85, "h": 12 },
      { "x": 320, "y": 185, "w": 75, "h": 12 },
      { "x": 420, "y": 235, "w": 80, "h": 12 },
      { "x": 520, "y": 190, "w": 95, "h": 12 },
      { "x": 560, "y": 140, "w": 70, "h": 12 }
    ]
  }
 ]
}
```


Notes (so it works even if your engine is strict)

- If your game **only supports** `theme.bg`, `theme.platform`, `theme.blob`, you can **delete** `accent`, `blobSymbol`, and `blobStyle` without breaking the rest.
- If you tell me what your renderer expects for “symbol” (emoji texture, sprite id, SVG path, etc.), I can format it exactly for your engine.”

My prompt:

“change the text to white in darker bacjgrounds”

ChatGPT 5.2 - output #2:

“Got it 👍 you want **text color to automatically switch to white on darker backgrounds**.

The clean way is to add a `text` (or `textColor`) field inside each level’s `theme`, and set it based on how dark the `bg` is.

Here’s your updated JSON with **white text on dark backgrounds** and **dark text on light backgrounds**:


```
{
  "schemaVersion": 1,
  "levels": [
    {
      "name": "Neon Alley",
      "gravity": 0.7,
      "jumpV": -11.2,
      "theme": {
        "bg": "#0B1020",
        "platform": "#2B3A67",
        "blob": "#00E5FF",
        "accent": "#FF2D95",
        "text": "#FFFFFF"
      },
      "blobStyle": "solid",
      "blobSymbol": "◆",
      "start": { "x": 70, "y": 240, "r": 24 },
      "platforms": [
```

```
    { "x": 0, "y": 336, "w": 640, "h": 36 },
    { "x": 90, "y": 270, "w": 120, "h": 12 },
    { "x": 250, "y": 220, "w": 90, "h": 12 },
    { "x": 380, "y": 170, "w": 140, "h": 12 },
    { "x": 560, "y": 250, "w": 70, "h": 12 }
  ]
},
{
  "name": "Candy Clouds",
  "gravity": 0.62,
  "jumpV": -10.6,
  "theme": {
    "bg": "#FFF1F7",
    "platform": "#FFC2D6",
    "blob": "#7C4DFF",
    "accent": "#00C853",
    "text": "#222222"
  },
  "blobStyle": "striped",
  "blobSymbol": "♥",
  "start": { "x": 60, "y": 210, "r": 26 },
  "platforms": [
    { "x": 0, "y": 334, "w": 640, "h": 36 },
    { "x": 70, "y": 260, "w": 110, "h": 12 },
    { "x": 220, "y": 200, "w": 120, "h": 12 },
    { "x": 390, "y": 240, "w": 80, "h": 12 },
```

```
    { "x": 500, "y": 170, "w": 120, "h": 12 },
    { "x": 590, "y": 260, "w": 50, "h": 12 }
  ]
},
{
  "name": "Forest Ruins",
  "gravity": 0.75,
  "jumpV": -12.0,
  "theme": {
    "bg": "#0F1F16",
    "platform": "#3D6B4F",
    "blob": "#FFD54F",
    "accent": "#A5D6A7",
    "text": "#FFFFFF"
  },
  "blobStyle": "glow",
  "blobSymbol": "☘",
  "start": { "x": 95, "y": 245, "r": 23 },
  "platforms": [
    { "x": 0, "y": 336, "w": 640, "h": 36 },
    { "x": 130, "y": 285, "w": 95, "h": 12 },
    { "x": 260, "y": 240, "w": 90, "h": 12 },
    { "x": 420, "y": 210, "w": 110, "h": 12 },
    { "x": 520, "y": 155, "w": 95, "h": 12 },
    { "x": 350, "y": 140, "w": 70, "h": 12 }
  ]
```

```
},
{
  "name": "Ice Circuit",
  "gravity": 0.58,
  "jumpV": -10.4,
  "theme": {
    "bg": "#E6F7FF",
    "platform": "#A7D8FF",
    "blob": "#1565C0",
    "accent": "#00B8D4",
    "text": "#1A1A1A"
  },
  "blobStyle": "outlined",
  "blobSymbol": "❄",
  "start": { "x": 80, "y": 230, "r": 25 },
  "platforms": [
    { "x": 0, "y": 334, "w": 640, "h": 36 },
    { "x": 120, "y": 270, "w": 80, "h": 12 },
    { "x": 240, "y": 230, "w": 85, "h": 12 },
    { "x": 360, "y": 255, "w": 90, "h": 12 },
    { "x": 470, "y": 210, "w": 110, "h": 12 },
    { "x": 560, "y": 170, "w": 70, "h": 12 }
  ]
},
{
  "name": "Volcano Vault",
```

```
    "gravity": 0.82,
    "jumpV": -12.6,
    "theme": {
      "bg": "#1B0B0B",
      "platform": "#6D2C2C",
      "blob": "#FF6D00",
      "accent": "#FFD600",
      "text": "#FFFFFF"
    },
    "blobStyle": "lava",
    "blobSymbol": "▲",
    "start": { "x": 65, "y": 245, "r": 22 },
    "platforms": [
      { "x": 0, "y": 336, "w": 640, "h": 36 },
      { "x": 80, "y": 290, "w": 100, "h": 12 },
      { "x": 210, "y": 235, "w": 85, "h": 12 },
      { "x": 320, "y": 185, "w": 75, "h": 12 },
      { "x": 420, "y": 235, "w": 80, "h": 12 },
      { "x": 520, "y": 190, "w": 95, "h": 12 },
      { "x": 560, "y": 140, "w": 70, "h": 12 }
    ]
  }
 ]
}
```


🔥 If you want it AUTOMATIC (better for your game)

Instead of hardcoding, you can calculate text color:

```
function getTextColor(bgHex) {
 const r = parseInt(bgHex.substr(1, 2), 16);
 const g = parseInt(bgHex.substr(3, 2), 16);
 const b = parseInt(bgHex.substr(5, 2), 16);

 const brightness = (r * 299 + g * 587 + b * 114) / 1000;

 return brightness > 150 ? "#000000" : "#FFFFFF";
}
```

Then:

```
theme.text = getTextColor(theme.bg);
```

If you want, I can also:

- add **moving platforms**
- add **spikes / hazards**
- add **goal flags**
- or make this into a **full P5.js game level system** (fits your CS106 project 👀)

Just tell me 👍”

My prompt:

“<!doctype html> <html lang="en"> <head> <meta charset="utf-8" /> <meta name="viewport" content="width=device-width, initial-scale=1" /> <title>Example 3 — JSON Levels + Level Class</title> <link rel="stylesheet" href="style.css" /> </head> <body> <script src="https://cdn.jsdelivr.net/npm/p5@1.9.0/lib/p5.min.js"></script> <script src="sketch.js"></script> </body> </html> why is this in error”

ChatGPT 5.2 - output #3:

“Your HTML is **not actually “wrong”**. Most “errors” people see with this exact file come from one of these issues:

1) Your file paths don’t exist

You link to:

- `style.css`
- `sketch.js`

If either file is missing or not in the same folder as this HTML, the browser will show **404 errors** (often in the console), and your page may look blank.

Fix
Make sure your folder looks like:

- `index.html`
- `style.css`
- `sketch.js`

(or change the paths to match your real names)

---

2) You opened the file without a local server

If you double click the HTML and open it as a `file://` URL, some setups (especially when loading JSON, images, or using `loadJSON`) will throw errors due to browser security rules.

Fix
Run a local server, for example:

- VS Code → “Live Server” extension
- or `python -m http.server` in that folder

Then open the `http://localhost...` link.

---

3) The p5 CDN is blocked or failing

If that CDN request fails, p5 won't load, and `sketch.js` will error because `createCanvas` etc won't exist.

Fix
Open DevTools Console and look for:

- "Failed to load resource … p5.min.js"
  If yes, try this CDN instead:

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.9.0/p5.min.js"></script>
```

---

4) Your actual error is inside `sketch.js`

Even if HTML is fine, a syntax error in `sketch.js` will show as "page broken."

Fix
Open DevTools Console and check the first red error line. It'll usually say something like:

- `Uncaught SyntaxError ... sketch.js:12`

Quick "safe" version of your HTML

This version removes the CSS link (in case that's the 404) and uses an alternate p5 CDN:

```
<!doctype html>

<html lang="en">

 <head>

   <meta charset="utf-8" />

   <meta name="viewport" content="width=device-width, initial-scale=1" />

   <title>Example 3 — JSON Levels + Level Class</title>

 </head>
```

```
 <body>
   <script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.9.0/p5.min.js"></script>
   <script src="./sketch.js"></script>
 </body>
</html>
```

If you paste the exact red console error text here

I can tell you **the exact cause** immediately (it’s almost always #1 or #4).”